

Neel Shah

AI ENGINEER · AZURE DATA SCIENTIST ASSOCIATE · DATABRICKS CERTIFIED DATA ENGINEER ASSOCIATE

Gujarat, India. (Willing to Relocate)

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Profile

AI Engineer with 2 years of hands-on experience building and shipping **production-grade AI systems**, from multi-agent frameworks and LLM-powered decision engines to large-scale computer vision pipelines. Delivered end-to-end AI solutions for **clients across USA, Saudi Arabia, Germany, and the Middle East**, spanning e-commerce, government, manufacturing, and automotive sectors. **Ranked Top 0.1%** (64th of 20,000+ teams) in the Amazon ML Challenge 2024 and a **5X Hackathon Winner**. Certified Microsoft Azure Data Scientist and Databricks Data Engineer. Thrives in fast-moving environments where owning problems end-to-end and shipping real impact matters.

Professional Experience

Neuramonks

Gujarat, India (Hybrid)

AI ENGINEER (INTERNSHIP → FULL-TIME)

Feb. 2025 - present

- Initially joined as an AI Engineer Intern and was converted to a full-time role after 3 months due to strong performance and significant contributions.
- Engineered an AI-powered **surveillance, workforce productivity monitoring, and person re-identification (ReID) platform** for diamond polishing facilities, enabling real-time employee tracking and working/non-working activity analysis across large-scale CCTV deployments on on-prem GPU infrastructure. Designed and optimized end-to-end computer vision pipelines for person detection, activity monitoring, and employee identification in dense industrial environments with varying lighting, occlusions, and overlapping movements. Experimented with and fine-tuned **YOLO, RF-DETR**, and other detection architectures for improved inference speed and detection accuracy, while benchmarking advanced tracking and ReID approaches including **BoT-SORT, ByteTrack, DeepSORT, AWS Rekognition**, and open-source embedding models to enhance surveillance analytics, employee identification reliability, and reduce ID switching during long-duration tracking scenarios.
- Architected a comprehensive end-to-end AI system for a **Saudi Arabian e-commerce platform with over 10k+ products**, integrating an intelligent chatbot for personalized product recommendations and engineering an **agentic AI system** for metadata tag generation from product images and descriptions, enabling real-time user query handling with dynamic product suggestions and follow-up filtering questions.
- Engineered a CrewAI-based multi-agent framework for the **Middle Eastern Governments**, demonstrating **99% concordance with expert human evaluators**, performing intelligent **RFP evaluation** by aligning RFP content with national EA standards and implementing **proposal scoring** using a custom point-based system to prioritize optimal vendor submissions based on alignment, coverage, and clarity.
- Deployed **Qwen2.5 VL 72B and Qwen3 32B** open-source models on-premise for enhanced security and compliance, replacing OpenAI PoC to meet stringent government deployment requirements while maintaining superior performance standards.
- Evaluated state-of-the-art native ASR models, including NVIDIA Parakeet-TDT-0.6B-v3 and Canary-1B-v2, specifically targeting high-throughput (RTFx) and minimizing sub-word error rate (WER) thresholds.
- Engineered an enterprise-grade AI-powered intelligent decision-making engine for supplier and inventory insights and PO approvals, seamlessly integrated with the ERP system of a large service provider, leveraging **500+ SQL tables** and deployed at scale for **100+ sub-tenant user organizations**.
- Evaluated and benchmarked state-of-the-art opensource voice synthesis models, including Coqui TTS (XTTS) and Chatterbox, to assess zero-shot cloning capabilities and acoustic quality.

Aviato Consulting

Australia (Remote)

AI ENGINEER (INTERNSHIP)

May. 2024 - Jun. 2024

- Architected an AI-powered virtual fashion try-on system for Australian retail clients, leveraging **deep learning and computer vision** to enable realistic garment visualization, empowering consumers with smarter, more confident purchase decisions.
- Engineered a **multi-agent** B2B customer interaction system using **LangChain** and **Neo4j** knowledge graphs on **GCP**, delivering actionable customer insights across fashion retail and CRM use cases.
- Built sophisticated personalized **recommendation engines** combining collaborative filtering and graph-based retrieval on **Neo4j**, significantly improving product discovery, user engagement, and satisfaction for diverse consumer audiences.
- Delivered production-ready AI solutions across multiple **Australian client** projects, receiving strong client feedback for technical quality and measurable improvements in user satisfaction and retail outcomes.

Charusat Learning And Development Club (CLDC)

Gujarat, India

AI DEVELOPER

May 2023 - Jun. 2023

- Spearheaded the engineering of a deep learning-based **botnet detection model using TensorFlow, Keras, and Python**, significantly enhancing cybersecurity by proactively identifying and neutralizing malicious botnet activities.
- Architected an advanced **emotion detection model**, leveraging deep learning techniques to accurately recognize and classify human emotions from video and audio data with high precision.

Skills

AI & ML	Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning
Generative AI	LLMs, LoRA, RAG, Reranking, Guardrails, Diffusion models, TTS (Text to Speech), STT (Speech to text)
Agentic Systems	CrewAI, LangGraph, LangChain, MCP (Model Context Protocol)
AI Coding Tools	Claude Code, Cursor, Codex
Programming	Python, SQL, Java
Frameworks & Libraries	PyTorch, TensorFlow, Keras, Hugging Face, Sklearn, Numpy, Pandas, Matplotlib, OpenCV, FastAPI, FastMCP, n8n
ML Architectures	Transformers (encoder-decoder, causal, autoencoders), Mixture-of-Experts (MoE), PPO, RLHF, Diffusion Models
MLOps	Git, Docker, Kubernetes, Kafka, MLflow, W&B, Apache Airflow, CI/CD, VLLM, Unsloth, TensorRT, DeepStream
Cloud Platforms	Azure, AWS, GCP, Databricks
Web Development	React.js, Node.js, HTML

Certifications

Stanford University	Machine Learning Specialization
Nvidia	Fundamentals of Accelerated Computing with CUDA Python
Microsoft	Certified Azure Data Scientist Associate
Databricks	Certified Data Engineer Associate
Huggingface	Deep Reinforcement Learning
DeepLearning.AI	TensorFlow Developer, Deep Learning, Generative AI with Large Language Models, Convolutional Neural Networks.
RedHat	Fundamentals of Redhat Enterprise Linux 9
Oracle	Oracle Certified Professional, Java SE 6 Programmer
NPTEL (IITs)	Deep Learning, Reinforcement Learning, Machine Learning, Data Base Management System, DSA using Java

Achievements

- Won **1st place** in **DataQuest 2024**, sponsored by **MSBC Group**, and secured **top 5 position** in the Healthathon organized by the I Am Patient organization and MSBC Group.
- Secured **3rd position** among 180 top-tier teams in **The Maverick Effect AI Challenge 2024**, supported by NASSCOM, and **2nd place** in the **IndoML Datathon 2024 (Phase 1)**, sponsored by Nieleniq, competing against 350+ teams.
- Secured **1st place** in one track of Phase 1 (2025) and **1st place** in both tracks of Phase 2 (2025) of the **IndoML Datathon**, competing among 400+ teams.
- Secured **Top 0.1%** out of 20,000+ teams in the **Amazon ML Challenge 2024**, utilizing 400+ GPU hours over three days, and achieved **All-India Top 0.3%** out of 23,000+ teams in the **Amazon ML Challenge 2025**.
- Selected as a **finalist** in the **AI Hackathon 2024** at the Indian Institute of Management (IIM), Ahmedabad, achieved **Top 11%** out of 196 teams in the **Meesho Visual Taxonomy Data Challenge CODS COMAD**, predicting product attributes from images.
- Led a team to **2nd position** at the University level in the **Smart India Hackathon 2023**, a nationwide initiative, and was among the **top 11 teams** selected for the **OpenACC Nvidia Hackathon** at IISc, Bangalore.
- Achieved **Top 5% nationwide** in the Database Management System course by **IIT Kharagpur** via NPTEL, and recognized as an **NPTEL Discipline Star** for completing over 50 weeks of IIT courses in computer science with a consistent score.
- Invited as a **Resource Person** by CSI & IEEE to deliver hands-on training on the MERN stack, and later invited by **IEEE CSPIT SB** to deliver a session on RAG, GANs, and YOLO.
- Appointed as **Student Representative** for the Board of Studies (AI & ML) (2023–2025), serving as a key liaison to refine curriculum and academic policies for the university.

Projects

Brain - TV

- Architected an innovative **AI neuroscience application** leveraging an **Encoder-Decoder architecture** with **ViT (Vision Transformer)** and **diffusion models** to visualize human thoughts from fMRI signals.
- Improved model performance by focusing on masked brain modeling and latent diffusion model conditioning using **fMRI latent data**.

MediPriority: AI-Driven Healthcare Management System

- Architected **MediPriority**, an AI-driven healthcare management system to streamline patient care from **pre-consultation to post-consultation**.
- Deployed **multi-agent AI systems**, incorporating text and video analysis agents, urgency scoring, and personalized health recommendations for enhanced patient outcomes.
- Integrated **dynamic urgency scoring**, **AI avatars for post-consultation support**, and **Google Calendar synchronization** to deliver seamless and efficient patient management.

PROTEAI – PROTEin-based AI for Therapeutic Drug Discovery

- Architected a **sequence-to-sequence (Seq2Seq) model** based on ByT5-base architecture to translate protein sequences into SMILES representations of potential drug molecules.
- Improved model robustness using a **byte-level tokenization strategy**, reducing preprocessing complexity compared to traditional methods.
- Explored multiple approaches, including character-level multi-class classification, to optimize the protein-to-compound prediction process.
- Contributed to AI-driven methodologies that accelerated therapeutic drug discovery in pharmaceutical research.

Education

Charotar University of Science and Technology

BACHELOR OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

9.6 CGPA

2022-2026